# IF EVERY OBJECT HAS A COVER BY QUASI-COMPACT OBJECTS THEN COMPACT OBJECTS ARE SUMMANDS OF FINITE COMPLEXES OF EXTENSION SHEAVES

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the proof is written out in full: the choice of the generating set, and the verification of the two conditions of the criterion with the reason for each.

*Editorial note.* The label lemma-compact-objects-if-enough-qc names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Editorial note.* The statement and proof are copied from the pinned chapter file; the three cited inputs are kept as external pointers with their canonical Stacks tags.

*Allowed standard prerequisites (editorial).*

- Compactness in terms of a generating set. Same chapter, the lemma describing compact objects through a set of generators satisfying two conditions. Supplies the criterion applied to the set of extension sheaves j_!O_U over quasi-compact U.
- Every module is a quotient of a direct sum of extension sheaves. Modules on Sites, the lemma representing an O-module as a quotient of a direct sum of j_!O_U. Supplies condition (1) of the criterion, together with the covering assumption.
- Sections commute with directed colimits. Sites, the lemma computing sections of a directed colimit as the colimit of the sections. Supplies condition (2) of the criterion through the identity Hom(j_!O_U, F) = F(U).

*Source reference.* Original reference 094B: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 13934–13934. Prerequisite pointer invoked at proof line(s) 14045; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0738: sources/stacks-chow/tags.txt, source lines 9052–9052. Prerequisite pointer invoked at proof line(s) 14056; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 03EW: sources/stacks-chow/tags.txt, source lines 4471–4471. Prerequisite pointer invoked at proof line(s) 14049; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Assume every object of $\mathcal{C}$ has a covering by quasi-compact objects. Then every compact object of $D(\mathcal{O})$ is a direct summand in $D(\mathcal{O})$ of a finite complex whose terms are finite direct sums of $\mathcal{O}$-modules of the form $j_!\mathcal{O}_U$ where $U$ is a quasi-compact object of $\mathcal{C}$.*

*Proof.* Apply Lemma 094B where $S \subset \mathrm{Ob}(\mathit{Mod}(\mathcal{O}))$ is the set of modules of the form $j_!\mathcal{O}_U$ with $U \in \mathrm{Ob}(\mathcal{C})$ quasi-compact. Assumption (1) holds by Modules on Sites, Lemma 03EW and the assumption that every $U$ can be covered by quasi-compact objects. Assumption (2) follows as

$$\mathrm{Hom}_{\mathcal{O}}(j_!\mathcal{O}_U, \mathcal{F}) = \mathcal{F}(U)$$

which commutes with direct sums by Sites, Lemma 0738. □